

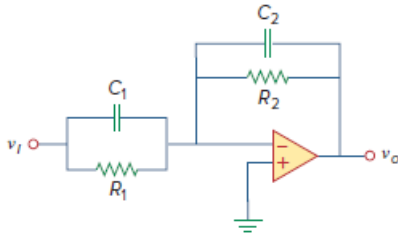
Problem

Design an op amp circuit, using Fig. 16.109, that will realize the following transfer function:

$$\frac{V_o(s)}{V_i(s)} = -\frac{s + 1000}{2(s + 4000)}$$

Choose $C_1 = 10 \mu\text{F}$; determine R_1 , R_2 , and C_2 .

Figure 16.109:



Step-by-step solution

Step 1 of 4

Consider the following op amp circuit diagram:

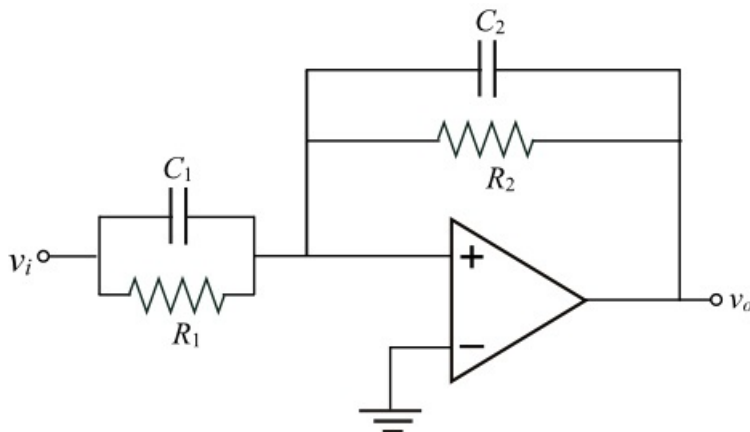


Figure 1: Op-amp circuit

Step 2 of 4

Apply Kirchhoff's current law at the noninverting terminal of the op amp in s-domain.

$$\frac{V_i(s) - 0}{R_1 \parallel \frac{1}{sC_1}} = \frac{0 - V_o(s)}{R_2 \parallel \frac{1}{sC_2}}$$

$$\frac{V_i(s)}{\left[\frac{R_1 \left(\frac{1}{sC_1} \right)}{R_1 + \left(\frac{1}{sC_1} \right)} \right]} = \frac{-V_o(s)}{\left[\frac{R_2 \left(\frac{1}{sC_2} \right)}{R_2 + \left(\frac{1}{sC_2} \right)} \right]}$$

$$\frac{V_i(s)}{\left(\frac{R_1}{sC_1 R_1 + 1} \right)} = \frac{-V_o(s)}{\left(\frac{R_2}{sC_2 R_2 + 1} \right)}$$

Rearrange the equation to get the transfer function.

$$\frac{V_o(s)}{V_i(s)} = - \frac{\left(\frac{R_2}{sC_2 R_2 + 1} \right)}{\left(\frac{R_1}{sC_1 R_1 + 1} \right)}$$

$$= - \frac{R_2 (sC_1 R_1 + 1)}{R_1 (sC_2 R_2 + 1)}$$

$$= - \frac{C_1 R_1 R_2 \left(s + \frac{1}{C_1 R_1} \right)}{C_2 R_2 R_1 \left(s + \frac{1}{C_2 R_2} \right)}$$

$$= - \frac{C_1 \left(s + \frac{1}{C_1 R_1} \right)}{C_2 \left(s + \frac{1}{C_2 R_2} \right)}$$

Substitute $-\frac{s+1000}{2(s+4000)}$ for $\frac{V_o(s)}{V_i(s)}$.

$$-\frac{s+1000}{2(s+4000)} = -\frac{C_1 \left(s + \frac{1}{C_1 R_1} \right)}{C_2 \left(s + \frac{1}{C_2 R_2} \right)}$$

Compare the equation notice that

$$\frac{C_1}{C_2} = \frac{1}{2}, \quad \frac{1}{C_1 R_1} = 1000, \quad \text{and} \quad \frac{1}{C_2 R_2} = 4000$$

Substitute 10×10^{-6} F for C_1 in equation $\frac{C_1}{C_2} = \frac{1}{2}$.

$$\frac{10 \times 10^{-6}}{C_2} = \frac{1}{2}$$

$$\Rightarrow C_2 = 20 \times 10^{-6} \text{ F}$$

$$\Rightarrow C_2 = 20 \mu\text{F}$$

Thus, the value of capacitance C_2 is $\boxed{20 \mu\text{F}}$.

Substitute 10×10^{-6} F for C_1 in equation $\frac{1}{C_1 R_1} = 1000$.

$$\frac{1}{(10 \times 10^{-6}) R_1} = 1000$$

$$\Rightarrow R_1 = \frac{1}{1000(10 \times 10^{-6})}$$

$$\Rightarrow R_1 = 100 \Omega$$

Thus, the value of resistance R_1 is $\boxed{100 \Omega}$.

Step 4 of 4

Substitute 20×10^{-6} F for C_2 in equation $\frac{1}{C_2 R_2} = 4000$.

$$\frac{1}{(20 \times 10^{-6}) R_2} = 4000$$

$$\Rightarrow \frac{1}{R_2} = 4000(20 \times 10^{-6})$$

$$\Rightarrow R_2 = \frac{1}{4000(20 \times 10^{-6})}$$

$$\Rightarrow R_2 = \frac{1}{4000(20 \times 10^{-6})}$$

$$R_2 = 12.5 \Omega$$

Thus, the value of resistance R_2 is $\boxed{12.5 \Omega}$.